

AI Application Development Bootcamp

Day 1: AI in Practice + First Application





Welcome and Course Introduction

Welcome to the AI Application Development Bootcamp.

- Over five weeks, build real AI-powered applications: chatbots, document assistants, MCP servers
- By the end, you'll have a portfolio of working projects to show employers

Today's plan:

- Morning: what generative AI is, how it works, real-world use, and its limits
- Afternoon: build your first chatbot, plus setting up your portfolio repo



Today's Lecture

- Setup and downloads (get the LLM model pulling now)
- What is generative AI? Core concepts and paradigms
- AI in practice: real-world applications and limits
- Popular AI models and how they work
- Limitations of AI models
- Responsible AI and data privacy (Singapore PDPA)
- Local AI and basic application architecture
- Wrap-up and afternoon lab preview



Your Git Repo Is Your Portfolio

- Hiring runs on proof of work now
 - Your resume lists skills; a public repo proves you can build
- Recruiters check your GitHub, not just your resume
 - Pinned projects, READMEs, commit history, contribution graph
- A public repo sells your skills around the clock
- Turn your 7+ bootcamp projects into one public repo
 - Graduate with a portfolio, not just a certificate



Set Up Your Portfolio Repo

- Create a public GitHub repo (use your DigiPen email for the free Student Pack)
- Give every project a README: what it does, why, tech, a screenshot
- Pin your 3 to 6 best projects; add a short profile README
- Commit as you go, with small, clear messages
- Never commit secrets: use a .gitignore, keep keys and tokens out
- Bonus: deploy your best project live (Streamlit Cloud, Vercel, Netlify)





Setup and Downloads

Getting Your Tools Ready



Step 1: Install Ollama

Ollama lets you run large language models locally. No internet, no cloud API, no usage fees.

- Go to <https://ollama.com> and download for your OS
- Run the installer and follow the prompts
- Open a terminal and verify:

```
ollama --version
```

Once verified, pull the model (~7 GB download):

```
ollama pull gemma4:e2b
```

Runs on 8 GB RAM, no GPU needed; ~7 GB download.



Step 2: Verify Your Model

Once the download finishes, test it:

```
ollama run gemma4:e2b  
>>> "Hello, tell me a joke"
```

If you get a response, your local AI is working. Type /bye to exit.

Fallback: if you have limited disk space, use:

```
ollama pull llama3.2:3b
```

Fallback model needs ~2 to 3 GB GPU VRAM (or 8 GB system RAM CPU-only); ~2 GB free disk.



Afternoon Lab: Tech Stack Preview

Role	Tool	What It Does
Brain (Local)	Ollama + Gemma 4 E2B	Runs the LLM on your machine
Brain (Cloud)	OpenAI / Anthropic API	Cloud LLM access via LiteLLM
Universal Plug	LiteLLM	Unified interface for any LLM
Memory	Mem0 + ChromaDB	Persistent conversation memory
Sensory	nomic-embed-text	Text embeddings
Skeleton	Python	Application logic
Face	Streamlit	Optional web UI

This afternoon you'll build a chatbot with these components.





What Is Generative AI?

The Big Picture





AI, Machine Learning, and Deep Learning

- **Artificial Intelligence:** Building systems that do things requiring human intelligence
- **Machine Learning:** Systems learn patterns from data instead of hand-coded rules
- **Deep Learning:** ML with neural networks (many layers), learns complex patterns
- **Generative AI:** A subset of deep learning that creates new content: text, images, code, audio, video



The Fundamental Purpose of Generative AI

Generative AI does two things:

- **Analysis:** Understanding, summarising, extracting, and reasoning about existing content
- **Creation:** Producing new content across multiple modes:
 - Text: emails, reports, translations, creative writing
 - Images: photographs, illustrations, design mockups
 - Code: functions, scripts, applications, bug fixes
 - Audio/Video: speech, music, clips, animations



How It Works: Neural Networks

A neural network is layers of connected nodes that process information.

Think of it like a factory assembly line: raw input enters, passes through processing stations (layers), each station detects certain patterns, and a finished output comes out.

Analogy:

If you saw thousands of cat photos, you'd say: whiskers, pointy ears, fur. A neural network does the same thing at a much more granular level, detecting edges, textures, and shapes, combining them layer by layer.



How It Works: Training

Training means showing a model enormous data so it learns patterns.

- LLMs read billions of pages of text: books, websites, code repos
- The model adjusts billions of parameters to get better at predicting the next word

Core insight:

Language models are next-word prediction engines that have gotten so good they appear to understand and reason.

Once trained, the model is frozen. It doesn't learn from your conversation.



The Transformer Architecture

Most modern generative AI is built on the Transformer (2017).

- **Key innovation:** The attention mechanism lets the model focus on the most relevant parts of its input

Example:

"The bank by the river was steep."

"Bank" could mean financial or riverbank. Attention weighs "river" and "steep" to pick the right meaning.



Generative AI vs. Other Paradigms

- **Generative AI:** Creates new content based on learned patterns (e.g., ChatGPT writing an essay, DALL-E generating images)
- **Discriminative Models:** Classifies or categorises existing data (e.g., spam filters, fraud detection)
- **Rule-Based Systems:** Follows predefined if-then rules (e.g., tax calculators, spell checkers)

Real applications often combine all three approaches.





AI in Practice

Real-World Applications



AI Across Industries

- **Software Development:** Code assistants (Copilot, Claude Code) generate code, write tests, review PRs. Developers still need to review AI output.
- **Business Workflows:** Drafting emails, summarising meetings, generating reports, translating documents at scale.
- **Customer Systems:** AI chatbots handle inquiries, give recommendations, resolve tickets 24/7.



Where Generative AI Works Well

- **Content creation at scale:** Drafting, translating, adapting content across formats and languages
- **Code assistance:** Writing boilerplate, generating tests, explaining complex code
- **Data analysis:** Processing large text volumes and pulling out key insights
- **Brainstorming and personalisation:** Generating options, exploring alternatives, creative problem-solving



Where Generative AI Falls Short

- **Factual accuracy:** Models produce plausible but completely wrong information (hallucinations)
- **Real-time data:** Knowledge cutoffs mean no access to live information without external tools
- **Precise calculations:** Language models are not calculators and make arithmetic errors
- **Regulated decisions:** Medical, legal, financial decisions need human accountability



Where Generative AI Falls Short (cont.)

- **Emotional intelligence:** AI simulates empathy but doesn't truly understand emotions
- **Novel reasoning:** Truly novel scientific breakthroughs are still out of reach

Bottom line: always pair AI output with human review, especially for high-stakes decisions.



AI in Singapore

- **Government push:** National AI Strategy 2.0 targets healthcare, education, finance, urban planning
- **Skills framework:** IMDA Skills Framework for Generative AI (what this course is aligned to)
- **Local examples:** DBS Bank (AI for customer service, fraud detection), GovTech (AI public services), plus startups across logistics and legal tech

Understanding AI is becoming a core employability requirement in Singapore.





Break

Check Your Ollama Download





Popular AI Models

And How They Work



Cloud Models: April 2026 (Bootcamp Run 1)

Model	Provider	Strengths
GPT-5.5	OpenAI	Strong agentic coding and reasoning; 1M context; first mainstream chatbot (ChatGPT, 2022)
Claude Opus 4.7	Anthropic	Most-used coding model: 87.6% SWE-bench Verified; 1M context
Gemini 3.1 Pro	Google	Strong reasoning; NotebookLM for audio podcasts; Nano Banana Pro tops image gen; 1M context
Qwen3-Max	Alibaba	Chinese frontier; strong on coding and multilingual; affordable API
GLM-5	Zhipu AI	Chinese frontier; agentic and reasoning; rivals top US labs on benchmarks
DeepSeek V4	DeepSeek	Chinese frontier; MoE 685B total / 37B active; 1M context; cost-efficient API



Cloud Models: July 2026 (Now)

Model	Provider	Strengths
GPT-5.5	OpenAI	Strong agentic coding and reasoning; 1M context; first mainstream chatbot (ChatGPT, 2022)
Claude Fable 5	Anthropic	Anthropic's most capable GA model; currently top-rated coding model; 1M context
Gemini 3.5 Pro	Google	Strong reasoning and multimodal; Nano Banana Pro leads image gen; 1M context
Qwen3.7-Max	Alibaba	Chinese frontier; agentic coding, 35-hour autonomous runs; 1M context; API-only
GLM-5.2	Zhipu AI	Chinese frontier; open weights (MIT); cost-effective coding; 1M context
DeepSeek V4	DeepSeek	Chinese frontier; MoE 1.6T total / 49B active; 1M context; open weights (MIT); cheap API



Open-Weight Models: April 2026 (Bootcamp Run 1)

Model	Provider	Strengths
Llama 4 Scout	Meta	Multimodal MoE (17B active/109B total); 10M token context; runs via Ollama
Gemma 4	Google	Strong reasoning, math, and code; 31B dense or 26B MoE (3.8B active); 256K context; Apache 2.0; runs via Ollama
DeepSeek-V3.2	DeepSeek	Strong reasoning and code; 685B total, 37B active (MoE)
Qwen 3.5	Alibaba	Excellent multilingual support; Dense and MoE variants, 0.6B to 235B parameters
Stable Diffusion 3.5	Stability AI	Open image generation; runs locally



Open-Weight Models: July 2026 (Now)

Model	Provider	Strengths
Nemotron 3 Ultra	NVIDIA	Strongest US open-weight; 550B MoE (55B active); 1M context; open (OpenMDW)
Gemma 4	Google	Best small-to-mid open-weight; 31B dense; 256K context; Apache 2.0; runs via Ollama
DeepSeek V4	DeepSeek	Tops open-weight leaderboards; MoE 1.6T total / 49B active; 1M context; open (MIT)
GLM-5.2	Zhipu AI	Leads open-weight reasoning (91% GPQA); 744B / 40B active; 1M context; open (MIT)
FLUX.2	Black Forest Labs	Open image generation; up to 4MP; runs locally



A Note on Keeping Up

This table will be out of date within weeks.

Case in point: you just saw it. Ten weeks ago GPT-5.5 was brand new. Today OpenAI has previewed GPT-5.6 (codename “Sol”, 26 June 2026), Anthropic has shipped Claude Fable 5, and nearly every row on the previous slide has changed.

New models, capabilities, and providers show up constantly. Staying current is part of your job.

- Follow key AI researchers on social media
- Subscribe to newsletters (The Batch, TLDR AI)
- Experiment with new models when they come out



How LLMs Work: Simplified

- **1. Tokenisation:** Your text is broken into tokens (word-pieces)
- **2. Embedding:** Each token becomes a vector capturing its meaning
- **3. Attention:** Transformer layers use attention to determine which tokens matter most to each other
- **4. Prediction:** The model predicts the most likely next token, picks one, repeats



A Quick Code Example

Calling a language model in six lines:

```
from litellm import completion
response = completion(
    model="ollama/gemma4:e2b",
    messages=[{"role": "user",
               "content": "Explain AI"}]
)

print(response.choices[0].message.content)
```



Cloud vs. Open-Weight Models

Aspect	Cloud API	Open-Weight
Examples	GPT-5.5, Claude Fable 5, Gemini 3.5	DeepSeek V4, GLM-5.2, Kimi K2.6
Runs on	Provider servers	Your hardware
Internet	Required	Works offline
Cost	Pay per query/token	Free after hardware (plus electricity)
Data privacy	Data sent to provider	Data stays local
Capability	Most capable models	Smaller but improving fast

Most organisations use both: cloud for complex work, local for privacy-sensitive tasks.



Context Windows

A context window is the max text a model can consider at once: its working memory.

- **Then:** A couple of years ago, 8,000 tokens was standard
- **Now:** Most cloud models support 1M+ tokens. Llama 4 Scout supports 10M. That's ~750,000 words.
- **Local models:** 128K to 10M tokens depending on model and hardware.



AI Vocabulary Quick Reference

Already covered

Attention: Lets each token weigh every other token in the input.

Context window: Max tokens the model can hold in working memory.

Embedding: Vector capturing the meaning of a token or piece of text.

Hallucination: Confident, fluent output that is factually wrong.

Knowledge cutoff: The end-date of the model's training data.

Token: A word-piece. Models read and write text in tokens.

Transformer: The neural-net architecture behind modern LLMs.

Vector: A list of numbers used to represent meaning.

New terms

Agents: Models that act in a loop: call tools, read results, repeat.

Fine-tuning: Training further on new data to specialise the model.

MCP: Model Context Protocol: standard way for models to call external tools.

Memory: External store the model reads across conversations.

MoE: Mixture of Experts: a subset of params activates per token.

Quantization: Compressing weights (e.g. Q4 = 4-bit) to fit less RAM.

RAG: Retrieval-Augmented Generation: look up text, then generate.

Vibe coding: Writing code by describing intent to an AI and shipping what it gives back.



The Engineering Disciplines

Prompt engineering: Crafting one model input to get one good output. The skill of 2022-23.

Context engineering: Deciding what fills the model's context window: instructions, examples, retrieved docs, tool definitions, prior turns. The core skill of 2024+, most production AI work is context engineering, not prompt engineering.

Harness engineering: Building the loop around the model: input validation, output parsing, tool calls, retries, fallbacks, evaluation. Turns a clever demo into a reliable product.

Agentic engineering: Orchestrating multi-step model behaviour: when to act, when to ask, how to recover. Builds on all three above. Day 12 focus of this bootcamp.





Limitations of AI Models

What AI Can't Do



Hallucinations and Bias

- **Hallucinations:** AI sometimes produces confident text that's factually wrong. It generates statistically likely words that diverge from reality.
 - Invented citations, made-up statistics, events that never happened
- **Bias:** Models learn from human data, which has biases. Over-representation of certain perspectives leads to stereotyping or knowledge gaps.



Knowledge Cutoff and Understanding

- **Knowledge Cutoff:** Models are trained on data up to a certain date. No knowledge of events after that without external tools.
- **No True Understanding:** Current AI recognises and reproduces patterns without genuine comprehension. It can write a poem about grief without having experienced it.

That's both the strength (generate about anything) and the weakness (no real judgment).





Responsible AI

Ethics and Data Privacy



Core Responsible AI Principles

- **Fairness:** AI systems shouldn't discriminate. Test for bias actively.
- **Transparency:** Users should know when they're interacting with AI.
- **Accountability:** Human oversight, clear ownership when things go wrong.
- **Privacy:** Respect privacy, collect only what's needed, follow the law.
- **Safety:** Test for harmful outputs, build guardrails against misuse.



What This Means for Developers

Responsible AI in practice:

- Always review AI outputs before they reach end users
- Test with diverse inputs to catch bias early
- Be upfront about what's AI-generated vs. human-written
- Build guardrails so your app can't produce harmful content
- Document what your AI system can and can't do





Singapore PDPA for AI

The Personal Data Protection Act governs handling of personal data.

- **Consent:** Get consent before collecting or using personal data
- **Purpose Limitation:** Data only for its collected purpose
- **Data Minimisation:** Collect only what you need
- **Accuracy:** Keep data accurate and current
- **Protection:** Implement proper security measures



Protecting Sensitive Information

Spot and protect: names, NRIC numbers, addresses, financial and health data.

Basic anonymisation techniques:

- **Data masking:** Replacing values with masked characters (e.g., S****567A)
- **Perturbation:** Adding noise so individuals can't be identified
- **Pseudonymisation:** Swapping real identifiers with fake ones





Local AI

And Application Architecture



Local vs. Cloud AI

Aspect	Local	Cloud
Privacy	Data stays on your machine	Data sent to provider
Cost	Free after hardware investment (plus electricity)	Pay per API call / token
Access	Works offline, no network round-trip	Needs internet connection
Performance	Limited by your hardware	Most capable models, enterprise GPUs
Setup	Install Ollama + download model	Just need an API key



Basic Architecture: Input, Model, Output

Every AI application follows a simple pattern:

- **1. Input:** User provides input (question, document, image). App processes it.
- **2. Model:** AI model generates a response. Local (Ollama) or cloud (OpenAI, Anthropic).
- **3. Output:** App formats and displays the response to the user.





Wrap-Up

Lab Preparation



What We Covered Today

- What generative AI is and how it differs from other approaches
- How models work: neural networks, training, transformers
- Where AI gets used in practice and where it falls short
- Major models (cloud and open-weight) and the evolving landscape
- Limitations: hallucinations, bias, cutoffs
- Responsible AI principles and Singapore's PDPA
- Local vs. cloud AI and Input-Model-Output architecture



Afternoon Lab Preview

Build your first portfolio project:

- **First activity (due today): create a public GitHub repo and build inside it**
- Connect to Gemma 4 E2B via LiteLLM
- Set up ChromaDB and Mem0 for long-term memory
- Use nomic-embed-text for embeddings
- Assemble logic in Python (chatbot.py)

Bonus: Pull a different local model like Qwen 3.5 4B (ollama pull qwen3.5:4b) and compare response quality and speed against Gemma 4 E2B.

Two tracks: CS background (more independence) or scripting background (step-by-step with Claude as tutor).





Key Takeaways and Day 2 Preview

Generative AI is powerful but not magic. It has real limits and demands responsible use.

Day 2: Prompt Engineering and Context Engineering

- How to control AI behaviour and improve output accuracy
- From crafting prompts to context engineering
- Hands-on with AI context & skill files

